



VEHICLE SERVICE BOOKING & TRACKING SYSTEM

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Abstract

- A backend-based system that centralizes vehicle servicing by enabling customers to register vehicles, select service centers, check slots, book appointments and track service progress. The backend manages bookings, job cards, technician allocation, inspections, spare parts, repairs, invoices, payments and history. Built with Node.js, Express.js, MySQL and REST APIs for secure, organized and transparent service management.

Problem Statement & Objectives

- Problem: Vehicle servicing often involves manual booking, unclear service status, scattered records and poor coordination.
Objective 1: Centralize service booking and tracking.
Objective 2: Manage technicians, inspections, repairs, inventory and billing.
Objective 3: Provide secure role-based access and REST APIs.
Scope: Customers, technicians, service staff and administrators.

Existing vs Proposed System

Feature	Existing System	Proposed System
Booking	Manual / scattered	Centralized online booking
Tracking	Limited visibility	Service progress tracking
Records	Separate records	Centralized database
Billing	Separate process	Integrated invoices & payments
Coordination	Manual communication	Role-based workflow

Feasibility Study

- Technical: Node.js, Express.js, Sequelize and MySQL provide the required backend platform.
Economic: Uses open-source development tools and software with low infrastructure cost.
Operational: Role-based modules support customers, technicians, service staff and administrators.
Schedule: Modular backend development can be completed within the semester timeline.

Results & Testing

- REST API modules planned for booking, jobs, inventory and billing.
- Role-based authentication and secure password handling.
- CRUD operations, validation and database integration are supported.

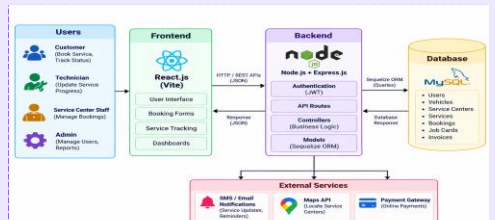
Conclusion

The system provides a centralized backend for vehicle service booking and tracking, connecting service operations, records and users through REST APIs and MySQL.

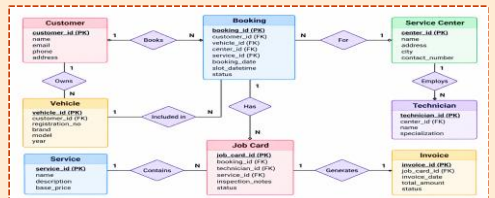
Tools & Technologies

- Backend: Node.js + Express.js + REST APIs
Database: MySQL + Sequelize ORM
Security: JWT authentication + bcrypt password hashing
Testing & Tools: Postman, VS Code, Git + GitHub
Services: Razorpay API, Leaflet + OpenStreetMap, Multer, Nodemailer

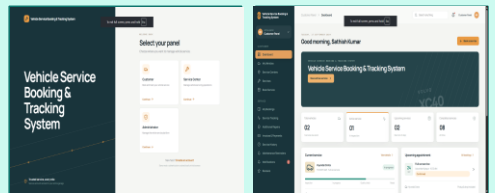
System Architecture



ER Diagram & Database Design



Implementation



Future Scope

- Mobile application integration
- Advanced service analytics and predictive maintenance
- Expanded notification and pickup/drop workflows